$\boxed{1}$

$\ln|x|$ $\quad x^{-2} \to \boxed{-x^{-1}}$

1.1) $\int_1^2 \left(\sqrt{x} + \frac{1}{x} + \frac{1}{x^2}\right) dx$

$x^{1/2} \to \frac{2x^{3/2}}{3}$

$= \left[\frac{2x\sqrt{x}}{3} + \ln x - \frac{1}{x}\right]_1^2 =$

$= \left(\frac{4\sqrt{2}}{3} + \ln 2 - \frac{1}{2}\right) - \underbrace{\left(\frac{2}{3} + 0 - 1\right)}_{-(-1/3)} = \frac{8\sqrt{2}-1}{6} + \ln(2)$

$\left(\frac{8\sqrt{2}-3}{6} + \ln 2\right) \quad \to +1/3 \to +2/6$

1.2) $\int_{-1}^2 \left(x^2 - 8x + x^3 + \sqrt[3]{x}\right) dx =$

$= \left[\frac{x^3}{3} - 4x^2 + \frac{x^4}{4} + \frac{3x\sqrt[3]{x}}{4}\right]_{-1}^2 =$

$= \left(\frac{8}{3} \overbrace{- 16 + \overbrace{\frac{16}{4}}^{4}}^{-12} + \frac{6\sqrt[3]{2}}{4}\right) - \left(-\frac{1}{3} \overbrace{- 4 + \overbrace{\frac{1}{4} + \frac{3}{4}}^{1}}^{-3}\right) =$

$= -6 - \frac{6\sqrt[3]{2}}{4} = -6\left(1 + \frac{\sqrt[3]{2}}{4}\right)$

1.3) $\int_1^2 \frac{6x^2+3x+4}{x} dx =$

$\to \int_1^2 6x\,dx + \int_1^2 3\,dx + 4\int_1^2 \frac{1}{x}\,dx$

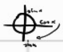
$= \left[3x^2 + 3x + 4\ln x\right]_1^2 =$

$= 9 + 3 + 4\ln 2 = \underline{12 + 4\ln 2}$

1.4) $\int_0^\pi (\cos x + \sin x)\, dx =$

$\to \int_0^\pi \cos x\,dx + \int_0^\pi \sin x\,dx \to \sin x - \cos x$

$= \left[\sin x - \cos x\right]_0^\pi = 0$



$\frac{d}{dx}\cos x = -\sin x$

1.5) $\int_{-2}^3 \left(\sqrt{|x|} - \sqrt[5]{x-1}\right) dx =$

$= \underbrace{\int_{-2}^3 \sqrt{|x|}\,dx}_{\downarrow} - \int_{-2}^3 \sqrt[5]{x-1}\,dx =$

$= \int_{-2}^0 \sqrt{-x}\,dx + \int_0^3 \sqrt{x}\,dx - \int_{-2}^1 \sqrt[5]{x-1}\,dx - \int_1^3 \sqrt[5]{x-1}\,dx$

$= \left[-\frac{3x\sqrt{-x}}{2}\right]_{-2}^0 + \left[\frac{3x\sqrt{x}}{2}\right]_0^3 - \left[\frac{5(x-1)\sqrt[5]{(x-1)}}{6}\right]_{-2}^3 =$

$= -\frac{6\sqrt{2}}{2} + \frac{9\sqrt{3}}{2} - \left(\frac{10\sqrt[5]{2}}{6} - \left(\frac{-15\sqrt[5]{-3}}{6}\right)\right) =$

$= \frac{9\sqrt{3}}{2} - \frac{6\sqrt{2}}{2} - \frac{10\sqrt[5]{2}}{6} + \frac{15\sqrt[5]{-3}}{6}$

1.6) $\int_0^{\pi/2} (\cos x + \arctan x)\, dx$

$\to \int_0^{\pi/2} \sin x\, dx + \int_0^{\pi/2} \arctan x\, dx =$

$\to \left[\arctan x \cdot x - \frac{1}{2} \cdot \ln(1+x^2)\right]_0^{\pi/2}$

$= \left[\cos x + \underline{\arctan x} \cdot x - \frac{1}{2} \cdot \ln(1+x^2)\right]_0^{\pi/2} =$

$= (0 - 1) + \left(\frac{\pi}{2}\arctan\frac{\pi}{2} - 0\right) + \left(-\frac{1}{2}\ln\left(1+\left(\frac{\pi}{2}\right)^2\right) + 0\right) =$

$= -1 + \frac{\pi}{2}\arctan\frac{\pi}{2} - \frac{1}{2}\ln\left(1+\left(\frac{\pi}{2}\right)^2\right)$

$\to \int f g' = f g' - \int f' g$

$\int \arctan x \cdot 1\, dx \qquad \arctan x \cdot x - \int \frac{1}{1+x^2} \cdot x\, dx$

$x \cdot \arctan x - \int \frac{x}{1+x^2}\, dx$

$1+x^2 = y \qquad 2x\,dx = dy \quad \to \int \frac{1}{2y}\,dy \to \frac{1}{2}\cdot\ln y \to \underline{\frac{1}{2}\ln|1+x^2|}$

$x \cdot \arctan x - \frac{1}{2}\ln(1+x^2)$

1.7) $\int_{-1}^{\sqrt{3}} \frac{5}{1+x^2}\, dx = \left[\frac{5}{2}\ln(1+x^2)\right]_{-1}^{\sqrt{3}} =$

$= \frac{5}{2}\ln(4) - \frac{5}{2}\ln(2) =$

$= 5\ln(2) - \frac{5}{2}\ln(2) =$

$= \frac{5}{2}\ln(2)$

1.8) $\int_0^1 \frac{x}{\sqrt{x+1}}\, dx \to \sqrt{x+1} = y \qquad y^2 - 1 = x \qquad dx = 2y\, dy$

$\to \int \frac{1}{\sqrt{x+1}} =$

$= \int_0^1 \frac{y^2-1}{\cancel{y}} \cdot 2\cancel{y}\, dy = \int_0^1 2y^2 - 2\, dy = \left[\frac{2}{3}y^3 - 2y\right]_0^1 =$

$= \frac{2}{3} - 2 = -\frac{4}{3}$

1.9) $\int_0^{2\pi} |\cos x|\, dx =$

$= \left[\sin x\right]_0^{2\pi} = 0$

1.10) $\int_0^3 \overbrace{\left|\overbrace{|x-2|}^{①} + 3\right|}^{②} dx$

① $\begin{cases} + \to \underline{x > 2} \\ - \to x < 2 \end{cases}$

② direi che si può levare sempre.

$\to \int_0^2 -x+5\, dx + \int_2^3 x+1\, dx =$

$= \left[-\frac{x^2}{2} + 5x\right]_0^2 + \left[\frac{x^2}{2} + x\right]_2^3 =$

$= (\overset{18}{-2+20}) + (\overset{7/5}{\frac{5}{2}+1}) = \boxed{97/5}$

1.11) $\int_0^5 \{x\}\, dx = \left[\frac{x^2}{2}\right]_0^5 = \frac{25}{2}$

1.12) $\int_{-2}^2 [x^2]\, dx = 2\int_0^2 x^2\, dx = 2\left[\frac{x^3}{3}\right]_0^2 = 2 \cdot \frac{8}{3} = \frac{16}{3}$

1.14) $\int_0^1 x \cdot e^{x^2}\, dx =$

$e^{x^2} = y \to x = \sqrt{\ln y} \to \frac{d}{dy}\sqrt{\ln y} = \frac{1}{2y\sqrt{\ln y}}$

$\frac{d}{dx} f^{1/2} = \frac{1}{2} f^{-1/2} \cdot f'$

$= \int_0^1 y \cdot \cancel{\sqrt{\ln y}} \cdot \frac{dy}{2\cancel{y\sqrt{\ln y}}} =$

$= \frac{1}{2}\int_0^1 dy =$

$= \frac{1}{2}\left[y\right]_0^1 = \frac{1}{2}\left[e^{x^2}\right]_0^1 = \frac{e-1}{2}$

$\int f g' = f g - \int f' g$

1.15) $\int_1^e x \log x\, dx = \frac{x^2}{2}\log x - \int_1^e \frac{1}{\cancel{x}} \cdot \frac{x^2}{2}\, dx =$

$= \frac{1}{2}\left(x^2 \log x - \int_1^e x\, dx\right) = \quad \to \frac{x^2}{2}$

$= \frac{1}{4}\left(2x^2\log x - e^2 + 1\right)$

1.16) $\int_0^{2\pi} \cos^2 x\, dx =$

$\cos x = y \to x = \cos^{-1}(y) \to dx = -\frac{1}{\sqrt{1-y^2}}\, dy$

$= \int_0^{2\pi} y^2 \cdot -\frac{dy}{\sqrt{1-y^2}} = \int_0^{2\pi} \frac{-y \cdot y}{\sqrt{1-y^2}}\, dy =$

$\sqrt{1-y^2} = v \to y = \sqrt{1-v^2}$

$\frac{1}{\cancel{2}\sqrt{1-y^2}} \cdot -\cancel{2}y = \frac{-y}{\sqrt{1-y^2}}\, dy = dv$

$= \int_0^{2\pi} \sqrt{1-v^2}\, dv$